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The development of the model of the atom

Atomic structure
Paper 1 · Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the first subatomic particle to be discovered.**	The electron	☐
2 ★	**Name the model of the atom with electrons embedded in a positively charged sphere.**	The plum pudding model	☐
3	**Name the model of the atom proposed after the nucleus was discovered.**	The nuclear model	☐
4	**Name the experiment that discovered the nucleus of an atom.**	The alpha particle scattering experiment	☐
5	**State what the alpha particle scattering experiment showed about the nucleus.**	The mass of the atom is concentrated in a very small, positively charged nucleus.	☐
6 ★	**Explain how the alpha particle scattering experiment showed that the nucleus is very small.**	Most alpha particles passed straight through; only a very few were scattered backwards.	☐
7 ★	**Explain how the alpha particle scattering experiment showed that the nucleus is positively charged.**	Some positive alpha particles were deflected or scattered backwards, repelled by the nucleus.	☐
8	**Name the model of the atom that came after the nuclear model.**	The Bohr model (Bohr adapted the nuclear model)	☐
9 ★	**Describe the Bohr model of the atom.**	A small positive nucleus with electrons orbiting at specific distances.	☐
10	**What were atoms thought to be before the electron was discovered?**	Tiny spheres that could not be divided.	☐
11	**Most alpha particles passed straight through the gold foil. State what this showed.**	Most of the atom is empty space.	☐

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	QUESTION	ANSWER	✓
12	**Explain why the alpha particle scattering experiment led to the plum pudding model being replaced.**	It could not explain why some alpha particles were deflected through large angles.	☐
13 ★	**Name the scientist who provided the evidence for the neutron.**	James Chadwick	☐
14	**Name the scientist who suggested that electrons orbit at specific distances.**	Niels Bohr	☐

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